

validate_mermaid_blocks.py — companion guide

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Overview

scripts/testing/validate_mermaid_blocks.py render-validates every ``mermaid block under a path using mmdc. A block that fails to render is a Constitution §11.4.168 defect — under the export pipeline it would ship as RAW SOURCE in the HTML/PDF instead of an image. This is the anti-bluff gate proving every diagram in the spec actually renders, and the enumerated-discovery tool (§11.4.118) for finding broken diagrams.

Prerequisites

- python3 (stdlib), mmdc (mermaid-cli), scripts/testing/mermaid.config.json

Usage

validate_mermaid_blocks.py [PATH]

- PATH — file or directory. Default: docs/research/mvp/final.

Output: one FAIL <file>:<line> [<type>] -> <error> per broken diagram, then mermaid-validate: total=N ok=M failed=K.

Exit: 0 all render · 1 ≥1 failed · 2 mmdc absent.

Common mermaid syntax faults (empirically confirmed in this spec)

1. **;** in a sequence/state message or note — mermaid treats `;` as a statement separator. Fix: `;` → `,` (or `split`). The validator stops at the *first* error per block, so a block may need several passes.
2. A second **:** in a stateDiagram-v2 transition label (first `:` is the label delimiter). Fix: reword, e.g. `:53` → `port 53`.
3. erDiagram attribute block inline on one line with `;`. Fix: one attribute per line inside `{ }`, format `TYPE name [PK|FK] ["comment"]`, no `;`. Note: `PK_FK` is invalid — use `PK, FK`.
4. `splitLineToFitWidth` does not support `newLines` on a `<br/>` label — if it persists, replace `<br/>` with a space.
5. Parens in a flowchart edge pipe-label (`|advertise()|`) — quote it: `|"advertise()"|`.

To debug one block: write it to `/tmp/x.mmd` and run `mmdc -i /tmp/x.mmd -o /tmp/x.png -b white -s3 -c scripts/testing/mermaid.config.json`.

Anti-bluff (§1.1)

A deliberately-broken diagram **MUST** be reported `failed=1` (exit 1); a good one `failed=0` (exit 0). Verified 2026-06-26.

Related scripts

- `scripts/testing/render_mermaid_blocks.py` — the renderer used by the export
- `scripts/testing/sync_all_markdown_exports.sh` — the export driver

Last verified: 2026-06-26 (full spec tree: 356 diagrams, failed=0).